

Pankaj Patel

Full Stack Engineer, Real-Time Communications and AI Infrastructure

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Professional Summary

Full Stack Engineer (SDE-3) with 3+ years of experience building production web products end to end across React, Next.js, TypeScript, Node.js, Python, AWS, and GCP, focused on real-time communication and AI infrastructure platforms. I have shipped billing, SIP calling, and real-time platform features that scaled from thousands to millions of live participants, and I have led a team of 5 engineers delivering full-stack applications.

Technical Skills

Frontend: React, Next.js, TypeScript, JavaScript, Redux, Zustand, Tailwind CSS, Material UI, responsive UI, real-time dashboards

Backend: Node.js, Express, Python (FastAPI), REST APIs, GraphQL, gRPC, WebSocket, Stripe billing, usage metering, RBAC

Cloud and DevOps: AWS (EC2, S3, Lambda), GCP, Docker, Kubernetes, GitLab CI/CD, Linux, monitoring and autoscaling

Data: MongoDB, PostgreSQL, Redis, BigQuery, aggregation pipelines, data modeling

Systems and Real-Time: WebRTC, SIP and VoIP, MediaSoup, NATS, Twilio SIP trunking, Go, C/C++, performance profiling

AI and LLM: LLM API integration, AI voice and chat agents, agent orchestration and tool calling, RAG and vector search

Practices: Full-stack development, software architecture, API design, distributed systems, CI/CD, code review, mentoring

Professional Experience

SDE-3, Full Stack Engineer, VideoSDK

Dec 2024 - Present

Surat, India | Real-time communication and AI agent platform

- Shipped a prepaid billing platform end to end using Stripe Payment Intents, subscriptions, React usage screens, and real-time metering; this moved the business from postpaid to upfront collection and eliminated uncollectible usage.
- Engineered SIP calling from scratch, including signaling, media negotiation, and call handling, making it the first audio and video SIP product at the company and enabling PSTN calling and AI voice agent revenue lines.
- Scaled signaling and media services from thousands to millions of live participants while maintaining zero customer-facing downtime during traffic spikes.
- Designed and launched an AI agent deployment platform in Python that let customers launch voice and chat agents into live rooms and phone calls with autoscaling, creating the foundation for the VideoSDK AI agent offering.
- Led the decommissioning of the legacy Archon media and signaling service, migrating 100% of traffic with zero downtime and reducing infrastructure costs while removing an entire on-call system.
- Launched two customer-facing features adopted by live-streaming clients: PK Battles for cross-room media forwarding, and room switching that moves participants between sessions with no reconnect.
- Created live React dashboards for bandwidth estimation and subscription telemetry that reduced network issue triage from hours to minutes for support and engineering teams.
- Resolved memory leaks, race conditions, and CPU bottlenecks across Node.js and C++ services and added health-check-driven autoscaling, improving uptime and reducing paged incidents at peak load.
- Developed core backend services in Node.js with NATS, Redis, and MongoDB, plus a multi-region session manager that self-heals dead sessions and custom Kurento C++ modules for DTMF and RTP splitting.

Full Stack Web Developer, CodeLink Infotech

Nov 2023 - Dec 2024

Surat, India | Software consultancy

- Led a team of 5 developers across MERN-stack client projects, scoping work, running code reviews, and mentoring junior engineers while delivering every project on schedule.
- Delivered an HRMS using Strapi CMS, MongoDB aggregation pipelines, and role-based dashboards that replaced manual HR workflows with self-serve reporting.
- Built a Hotel Management System in Next.js with a custom component library and embedded WebRTC chat and video, shipped as the primary booking and operations product for the client.

Projects

- Stock Market Analytics Platform** (React, Material UI, WebSocket): streaming charts and market data rendered live with sub-second updates.
- Linux Kernel Device Driver** (C, ioctl): character driver with a circular queue passing data between kernel and user space.

Education

B.Tech, Computer Engineering, Bhagwan Mahavir University | GPA: 8.8 / 10.0

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